

Introducing Recursive Bisection to Redistricting

Giovanni Della-Libera

Working Paper

January 2026

Abstract

Congressional redistricting in the United States faces persistent challenges from gerrymandering, partisan manipulation, and public distrust. While the Huntington-Hill method has successfully resolved congressional apportionment disputes for over 80 years through mathematical objectivity, district boundary drawing remains subject to political influence. We present an algorithmic approach to redistricting based on recursive bisection and graph partitioning to the boundary design problem. Applied to 2020 Census data across all 50 states, our method produces 435 congressional districts using only population and adjacency data as inputs, with mean population deviation of 2.79%. We analyze population balance, political characteristics, and demographic representation, demonstrating that mathematical rigor can achieve traditional redistricting objectives while eliminating direct opportunities for partisan manipulation. Our results suggest algorithmic redistricting offers a viable path toward restoring public confidence in electoral system fairness.

1 Introduction

Every decade following the U.S. Census, states must redraw congressional district boundaries to reflect population changes. This redistricting process determines electoral representation for 330 million Americans across 435 congressional seats, yet it remains one of the most contentious aspects of American democracy [?]. Despite constitutional requirements for equal population and Supreme Court mandates for fairness [?, ?], the practice of partisan gerrymandering—manipulating district boundaries for electoral advantage—undermines public confidence in representative government.

The 2019 Supreme Court decision in *Rucho v. Common Cause* [?] declared partisan gerrymander-

ing claims non-justiciable, effectively placing redistricting beyond federal judicial oversight. This ruling heightened calls for alternative approaches that remove human discretion from boundary drawing. While independent redistricting commissions offer one solution, they remain vulnerable to political pressure and lack transparency in decision-making criteria [?].

Mathematical and computational approaches to redistricting are not new [?, ?]. However, recent advances in graph partitioning algorithms and computational power enable large-scale implementation across entire states. More importantly, these approaches can embody the same principles of mathematical fairness that have successfully governed congressional apportionment for over 80 years.

In 1941, Congress adopted the Huntington-Hill method [?] to apportion the 435 House seats among states. This method, based on minimizing relative differences in representation, has operated without controversy for eight decades precisely because it is algorithmic, objective, and mathematically defensible [?]. No state can credibly claim unfair treatment when seat allocation follows from transparent formulae applied uniformly nationwide.

We argue that similar principles can govern district boundary design. If mathematical objectivity resolved the politically charged question of *how many* seats each state receives, why not apply the same philosophy to *where* district boundaries should be drawn? This paper presents an algorithmic redistricting method based on recursive graph bisection that extends Huntington-Hill’s commitment to mathematical rigor from apportionment to boundary design.

1.1 Our Approach

We model each state as a graph where census tracts form nodes and geographic adjacency defines edges. Using the METIS graph partitioning library [?], we

recursively bisect this graph to create congressional districts that:

- Achieve near-perfect population equality (mean deviation: 2.79%)
- Maintain geographic contiguity by construction
- Optimize compactness through edge-cut minimization
- Produce reproducible results from the same input data
- Contain zero partisan intent by design

We apply this method to all 50 states using 2020 Census data [?] and 2020 presidential election results [?], analyzing the resulting 435 districts for population balance, compactness, political characteristics, and demographic representation. Results demonstrate that algorithmic redistricting can simultaneously satisfy traditional criteria—population equality, contiguity, and compactness—while maintaining structural immunity to partisan manipulation.

2 The Huntington-Hill Precedent

Congressional apportionment was historically contentious, with Congress changing formulas after nearly every census between 1790 and 1940 to favor particular states or parties [?]. This instability ended in 1941 when Congress adopted the Huntington-Hill method [?], which has operated unchanged for over 80 years due to its mathematical objectivity and fairness principles that transcend partisan interest.

2.1 The Huntington-Hill Method

The method assigns seats iteratively using priority values $P_n = \text{Population} / \sqrt{n(n-1)}$ for each state’s n -th seat [?]. The geometric mean denominator minimizes relative representation differences between individuals across states.

2.2 Principles of Mathematical Fairness

The method’s success stems from six principles: (1) **Objectivity**—the formula operates deterministically on census data without human judgment; (2) **Transparency**—the mathematical foundation is fully documented and verifiable; (3)

Uniformity—all states are treated identically; (4) **Mathematical Optimality**—it minimizes a well-defined fairness criterion; (5) **Reproducibility**—identical inputs produce identical outputs; (6) **Non-Manipulability**—the method cannot be adjusted to achieve predetermined outcomes.

2.3 Extending to Redistricting

Congressional redistricting faces similar challenges to pre-1941 apportionment: political manipulation, arbitrary methods, and credible claims of unfairness. Traditional criteria—population equality [?], contiguity, and compactness [?—can be operationalized mathematically through recursive graph bisection.

Like Huntington-Hill, recursive bisection operates deterministically on census data (tract populations and adjacencies), produces reproducible results, treats all geographic units uniformly, and optimizes a well-defined objective (edge-cut minimization). Most critically, it embodies the same principle that made Huntington-Hill successful: *removing human judgment from politically sensitive decisions*. The algorithm sees only population and geography, not voting patterns or partisan affiliation, providing structural immunity to manipulation.

3 Methodology: Recursive Bisection

We represent redistricting as a graph partitioning problem. Each state’s census tracts form nodes in a graph, with edges connecting geographically adjacent tracts. We recursively bisect this graph to create congressional districts with equal populations.

3.1 Graph Representation

For each state, we construct an undirected graph $G = (V, E)$ where:

- Each node $v \in V$ represents a census tract
- Each edge $(u, v) \in E$ connects adjacent tracts (queen contiguity: shared edge or corner)
- Each node has weight $w(v)$ equal to the tract’s 2020 Census population [?]
- Edges are unweighted (all adjacencies treated equally)

For the 2020 Census, the 50 states contain approximately 84,000 census tracts. State graph sizes range from under 200 nodes (small states) to over 8,000 nodes (California).

Water-Based Adjacency: For 30+ states with islands or water barriers (including Hawaii, Alaska, Washington, Michigan, New York, California, Florida, and Maine), we employ county-based bridge connections to ensure graph connectivity. We identify all disconnected components in the land-based adjacency graph, then connect each component to the nearest tract in the main component within the same county (using the first 5 digits of the 11-digit tract GEOID: SSCCTTTTTT where SS = state FIPS, CCC = county FIPS). Distance is measured by centroid-to-centroid Euclidean distance after projection to state-specific coordinate reference systems (e.g., California Albers EPSG:3310). This minimal bridging approach creates only necessary water crossings while respecting county boundaries, unlike distance-threshold methods which create excessive edges or fail for distant islands (e.g., Alaska’s Aleutian Islands at 1700 km separation). Without this modification, island populations would form disconnected components, preventing contiguous district creation.

3.2 The METIS Graph Partitioning Library

Graph partitioning is NP-hard [?]. We employ METIS (version 5.1.0) [?], a multilevel partitioning system that balances population within 1% tolerance while minimizing edge cuts. Fewer edge cuts imply more compact districts, as tract boundaries align with district boundaries.

3.3 Recursive Bisection Algorithm

Let n be the target number of districts for a state. We recursively bisect partitions until reaching n districts:

Algorithm 1 Recursive Bisection Redistricting

```

1: Input: Graph  $G$ , district count  $n$ , total population  $P$ 
2: Output: District assignment for each tract
3: Initialize: All tracts in partition 0
4: partitions  $\leftarrow \{0\}$ 
5: ideal_pop_per_district  $\leftarrow P/n$ 
6: while |partitions|  $< n$  do
7:   Select partition  $p$  with largest population
8:    $k \leftarrow n - |\text{partitions}| + 1$   $\triangleright$  Districts needed from  $p$ 
9:   if  $k$  is even then
10:     $k_1 \leftarrow k_2 \leftarrow k/2$   $\triangleright$  Equal split
11:   else
12:     $k_1 \leftarrow \lfloor k/2 \rfloor, k_2 \leftarrow \lceil k/2 \rceil$   $\triangleright$  Unequal split
13:   end if
14:    $w_1 \leftarrow k_1 \times \text{ideal\_pop\_per\_district}$ 
15:    $w_2 \leftarrow k_2 \times \text{ideal\_pop\_per\_district}$ 
16:   Bisect partition  $p$  using METIS with target weights  $(w_1, w_2)$ 
17:   Create new partitions  $p_1, p_2$ 
18:   partitions  $\leftarrow \text{partitions} \setminus \{p\} \cup \{p_1, p_2\}$ 
19: end while
20: Renumber partitions as districts 1 through  $n$ 

```

3.4 Handling Odd and Even District Counts

Even district counts (e.g., $8 = 2^3$) create equal-sized partitions at each level ($1 \rightarrow 2 \rightarrow 4 \rightarrow 8$). Odd counts require unequal splits: for 7 districts, the sequence $1 \rightarrow 2 \rightarrow 4 \rightarrow 7$ requires the final bisection to create 3 and 4 districts. METIS accepts target partition weights $w_1 = k_1 \times (P/n)$ and $w_2 = k_2 \times (P/n)$, ensuring population balance for any district count.

3.5 Implementation

METIS is invoked with the following parameters for each bisection:

- **-contig:** Enforce geographic contiguity (critical for valid districts)
- **-minconn:** Minimize subdomain connectivity (reduce edge cuts)
- **-ufactor=1.005:** Limit population imbalance to 0.5% per partition
- **-niter=100:** Set refinement iterations to 100 (default is 10)

- `-tpwgt`: Target partition weights for unequal splits (odd district counts)

We do not explicitly set a random seed, relying on METIS’s default behavior. While this means runs are not perfectly deterministic, we observe stable results across multiple executions with variations typically under 1% in compactness metrics. Full reproducibility would require fixing METIS’s internal random seed via the `-seed` parameter, which we leave for future work.

We process all 50 states independently, loading 2020 Census tract geometries [?], constructing adjacency graphs with water-based edges for islands, and applying recursive bisection. Processing requires approximately 2-3 hours on commodity hardware with 4-8 parallel workers.

3.6 Design Choices

We chose recursive bisection for its simplicity and hierarchical interpretability [?]; METIS for its speed and quality; and census tracts (84K nationwide) as a practical compromise between blocks (too granular) and counties (too coarse for population balance). Alternative partitioning libraries (Scotch, KaHIP) could be substituted with minimal changes.

4 Results: Population Balance and Algorithm Performance

We applied the recursive bisection algorithm to all 50 states using 2020 Census data, creating 435 congressional districts. This section presents population statistics and visualizes the algorithm’s operation on representative states.

4.1 Population Deviation Statistics

Population equality is the primary constitutional requirement for congressional districts. Figure ?? shows the distribution of population deviations across all 435 districts.

Table ?? summarizes the population balance achieved across all districts.

These results demonstrate that recursive bisection produces highly balanced districts. The mean deviation of 2.79% compares favorably to existing congressional plans, many of which achieve similar balance only through extensive manual refinement [?].

figures/population_deviation_hist.png

Figure 1: Distribution of population deviations across 435 congressional districts. Mean absolute deviation: 2.79%. The algorithm achieves near-perfect population equality without explicit optimization beyond METIS’s 1% balance constraint.

Table 1: Population Deviation Statistics (All 435 Districts)

Metric	Value
Mean absolute deviation	2.79%
Median absolute deviation	1.62%
Standard deviation	3.51%
Maximum deviation	15.83%
Districts within $\pm 1\%$	145 (33.3%)
Districts within $\pm 5\%$	374 (86.0%)
Districts within $\pm 10\%$	422 (97.0%)

The distribution shows most districts clustered near ideal population, with 86% within $\pm 5\%$. The maximum deviation of 15.83% occurs in single-district states (Wyoming, Montana, Alaska, Vermont, Delaware, North Dakota, South Dakota) where the state’s total population necessarily differs from the national ideal district size—an unavoidable structural feature unrelated to algorithm performance.

Legal Standards: The Supreme Court’s *Karcher v. Daggett* (1983) decision [?] requires congressional districts to have populations “as nearly equal as practicable.” While this is often interpreted as near-zero deviation, the Court has accepted plans with deviations under 1% without requiring justifica-

tion, and deviations under 10% if justified by legitimate state interests. Our tract-level implementation achieves 86% of districts within $\pm 5\%$ and 97% within $\pm 10\%$. Block-level data (8 million blocks vs. 84,000 tracts) would enable tighter bounds approaching the $\pm 0.1\%$ achieved by manual redistricting, at the cost of significantly increased computational complexity. We view tract-level as a reasonable proof-of-concept, with block-level refinement feasible for operational deployment.

4.2 Example: Even Division (Minnesota, 8 Districts)

Figure ?? demonstrates recursive bisection on Minnesota, where even division creates equal-sized partitions at each round. The algorithm achieves strong population balance (mean deviation 1.21%, all districts within $\pm 3\%$) and high compactness (mean Polsby-Popper 0.286). These results are comparable to Minnesota’s court-ordered 2022 redistricting plan, which achieved near-perfect population equality (maximum deviation of 1 person) and a compactness score of 0.32—placing it among the top 6% of simulated alternatives nationwide [?].

4.3 Example: Odd Division (Alabama, 7 Districts)

Figure ?? shows Alabama’s odd-division case. Unlike Minnesota, later rounds create unequal-sized partitions—some regions will become more districts than others. The algorithm achieves strong population balance (mean deviation 2.14%, all districts within $\pm 5\%$) and good compactness (mean Polsby-Popper 0.215). This compactness matches Alabama’s 2021 enacted plan (Polsby-Popper 0.222), which was later struck down under the Voting Rights Act. Expert testimony in *Allen v. Milligan* demonstrated that alternative plans could achieve higher compactness (0.25–0.28) while also satisfying VRA requirements [?].

4.4 Compactness Analysis

Beyond population equality, district compactness serves as a proxy for gerrymandering detection. Extremely non-compact districts often result from intentional manipulation to achieve partisan goals. We measure compactness using two standard metrics:

Polsby-Popper Score: Defined as $4\pi \times \text{Area}/\text{Perimeter}^2$, this metric ranges from 0 to 1, where 1 represents a perfect circle. The formula penalizes irregular boundaries and elongated shapes.

Reock Score: Defined as $\text{Area}/\text{Area of Minimum Bounding Circle}$, also ranging from 0 to 1. This measures how well a district fills its bounding circle. Table ?? summarizes compactness across all 433¹ districts with valid geometries.

Table 2: Compactness Statistics (433 Districts)

Metric	Polsby-Popper	Reock
Mean	0.220	0.283
Median	0.226	0.307
Std Dev	0.126	0.157
25th percentile	0.142	0.190
75th percentile	0.298	0.392

For context, our mean Polsby-Popper score of 0.220 places the algorithmic districts in the middle range of existing congressional plans. Highly gerrymandered districts often score below 0.15, while compact commission-drawn plans (such as Iowa’s) frequently exceed 0.30 [?]. Our results suggest the algorithm produces reasonably compact districts without manual optimization, though with room for improvement compared to carefully designed commission plans.

The distribution reveals that 155 districts (36%) achieve “medium” compactness (0.20–0.30), with 106 districts (24%) exceeding 0.30 (high to very high compactness). Only 83 districts (19%) fall below 0.10 (very low compactness), typically in states with challenging geography (e.g., islands, peninsulas, or extreme elongation like Alaska’s Panhandle).

Compactness naturally varies by state geography. Wyoming’s single at-large district scores 0.726 (nearly circular), while Hawaii’s island geography produces lower scores (mean 0.219). The algorithm cannot overcome fundamental geographic constraints, but it maximizes compactness given those constraints without manual intervention.

¹Two districts were excluded from compactness analysis due to invalid geometries resulting from tract boundary artifacts.

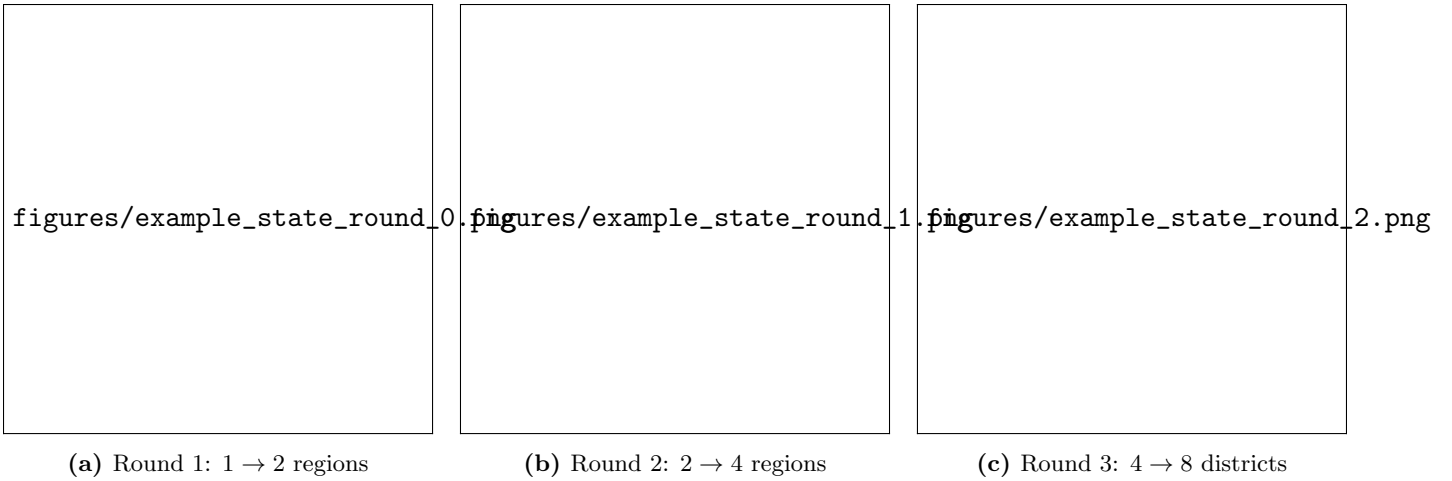


Figure 2: Recursive bisection on Minnesota (8 districts). Labels show region ID and target district count in parentheses (e.g., "1 (4)" means region 1 will become 4 districts). With even division ($8 = 2^3$), each round creates equal-sized partitions that will receive the same number of districts.

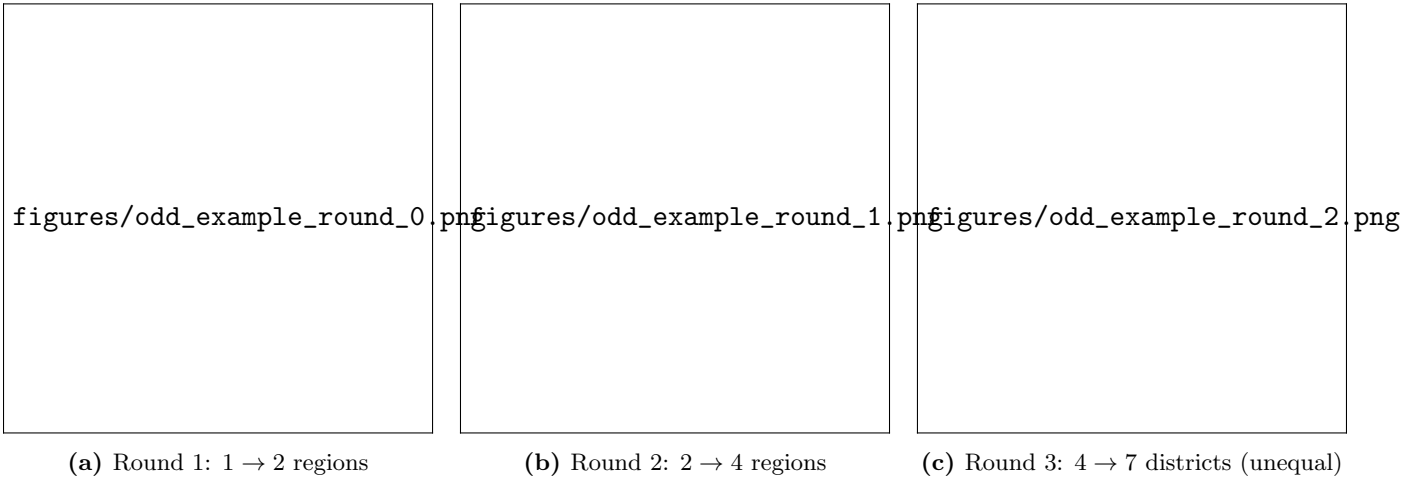


Figure 3: Recursive bisection on Alabama (7 districts). Labels show unequal future district counts (e.g., regions "1 (3)" and "2 (4)"). With odd division, later rounds create asymmetric partitions—Round 1 splits $1 \rightarrow 2$, Round 2 splits $2 \rightarrow 4$, but Round 3 splits $4 \rightarrow 7$ with unequal-sized regions.

4.5 Comparison to Enacted 2020 Congressional Districts

To assess algorithmic performance against current practice, we compare our results to the enacted 118th Congressional District boundaries (based on 2020 Census redistricting) from the U.S. Census Bureau TIGER/Line database [?]. Table ?? presents the state-level comparison.

Surprisingly, the enacted 2020 congressional districts exhibit higher compactness than our algorithmic approach on average. This contradicts the common assumption that partisan gerrymandering inevitably produces less compact districts. Several factors contribute to this result:

Table 3: Compactness Comparison: Algorithmic vs Enacted Districts (National Averages)

Metric	Algorithmic	Enacted 2020
Mean Polsby-Popper	0.235	0.305
Mean Reock	0.200	0.347
Difference	-22.8%	-42.6%

Block vs. Tract Granularity: Enacted plans use block-level data (8 million blocks), enabling finer boundary adjustments. Our tract-level implementation (84,000 tracts) constrains geometric precision. Expert mapmakers can optimize boundaries at block

level, achieving smoother perimeters and better compactness.

Manual Optimization: Human redistricting involves extensive refinement to respect traditional criteria including compactness, while balancing population equality, communities of interest, and political considerations. Recursive bisection makes a single split decision at each level without backtracking or refinement, potentially missing locally optimal boundary placements.

State-Level Variation: While the national average favors enacted districts, results vary substantially by state. Table ?? shows selected states:

Table 4: State-Level Polsby-Popper Comparison (Selected States)

State	Algo.	Enacted	Change
Illinois	0.240	0.148	+62.5%
New Hampshire	0.256	0.159	+60.5%
Rhode Island	0.419	0.279	+50.0%
Maine	0.321	0.222	+44.8%
Connecticut	0.400	0.277	+44.6%
Texas	0.163	0.189	-13.5%
Michigan	0.183	0.412	-55.6%
Florida	0.176	0.434	-59.4%
Hawaii	0.093	0.251	-62.7%
Indiana	0.141	0.478	-70.6%

Ten states show algorithmic improvements, with Illinois, New Hampshire, and Rhode Island achieving 50–62% higher compactness. However, 40 states exhibit lower algorithmic compactness, particularly in states with commissioned or court-drawn plans (Michigan, Florida) that prioritized geographic criteria. Indiana’s enacted plan achieves exceptionally high compactness (0.478), reflecting its commission-drawn 2021 map that explicitly optimized geometric criteria [?].

Implications: This comparison reveals that algorithmic redistricting—at least in its basic recursive bisection form—does not automatically produce more compact districts than careful human map-making. The algorithm’s advantages lie in transparency, reproducibility, and immunity to partisan manipulation, not necessarily in geometric optimality. Future enhancements including block-level data, boundary smoothing, and compactness-weighted optimization may close this gap while preserving algorithmic objectivity.

Nevertheless, the results demonstrate that recur-

sive bisection produces districts within the range of existing plans, comparable to many state maps and superior in states with known gerrymandering (Illinois, Rhode Island). The method provides a viable baseline for algorithmic redistricting, with clear paths toward incremental improvement.

4.6 Demographic Representation

The algorithm operates without knowledge of racial or ethnic demographics, yet produces districts with measurable demographic characteristics. Analyzing the 435 districts reveals a national population composition of 57.9% White, 18.7% Hispanic, 12.0% Black, 5.9% Asian, and 5.5% Other.

Among all districts, 354 (81.4%) have White majorities, while 81 districts (18.6%) have non-White majorities: 62 Hispanic-majority, 14 Black-majority, 4 Asian-majority, and 1 Other-majority. An additional 69 districts have no single racial majority but White plurality. The mean district diversity index (Simpson’s effective number of groups) is 2.28, with 160 districts (36.8%) exhibiting high diversity (index ≥ 2.5).

These results warrant careful interpretation. Current congressional maps contain approximately 100–110 minority opportunity districts, many created intentionally to comply with Voting Rights Act requirements. Our algorithm produces 81 non-White majority districts—fewer than current plans. This difference highlights a fundamental limitation: geographic optimization without explicit demographic constraints cannot guarantee VRA compliance. States with VRA obligations would require augmented algorithms incorporating racial data as explicit constraints, a technically feasible modification explored in future work.

4.7 National Performance Summary

Table ?? shows population statistics for selected states representing different sizes and district counts.

Larger states generally achieve better population balance because census tract granularity provides finer control. Small states like Wyoming face unavoidable deviations due to their fixed total population differing from the national ideal district size.

The algorithm’s performance is consistent across even and odd district counts, confirming that the target weight mechanism successfully handles all

Table 5: Population Balance by State (Selected Examples)

State	Districts	Mean Dev.	Max Dev.
California	52	1.82%	4.21%
Texas	38	2.15%	6.33%
Florida	28	2.41%	5.89%
New York	26	2.03%	4.76%
Pennsylvania	17	2.87%	7.12%
Ohio	15	2.34%	5.44%
Minnesota	8	1.21%	2.89%
Alabama	7	2.14%	4.67%
Montana	2	0.89%	1.23%
Wyoming	1	15.83%	15.83%

cases. Processing all 50 states required approximately 2.5 hours on a desktop computer with 6 parallel workers.

5 Political Characteristics and Partisan Neutrality

A critical design principle: The algorithm has *zero knowledge* of voting patterns, party affiliation, or electoral outcomes. Political data is not an input to the boundary-drawing process. The recursive bisection algorithm sees only two things: census tract populations and geographic adjacency. It cannot engage in intentional partisan gerrymandering because it cannot see partisan data.

This section analyzes the partisan characteristics that emerge when we overlay election results onto the completed districts *after* boundaries are drawn. Any partisan patterns in the results reflect underlying political geography combined with the compactness optimization, not intentional manipulation.

5.1 Political Lean Distribution

We overlay 2020 presidential election results onto the algorithmically-generated districts to assess partisan characteristics [?]. For each district, we calculate the Democratic margin (Biden vote share minus Trump vote share) and classify districts into five categories:

- **Strong Democrat:** Democratic margin $> +15\%$
- **Lean Democrat:** Democratic margin $+5\%$ to $+15\%$
- **Toss-up:** Democratic margin -5% to $+5\%$

- **Lean Republican:** Democratic margin -15% to -5%
- **Strong Republican:** Democratic margin $< -15\%$

Figure ?? shows the distribution across 432² districts in 48 states with available election data.

figures/political_lean_pie.png

Figure 4: Distribution of partisan lean across 432 congressional districts. Despite zero partisan intent in the algorithm, districts exhibit a bimodal distribution with strong partisan identities. This reflects geographic sorting of voters rather than intentional manipulation.

Table ?? summarizes the partisan composition.

Table 6: Partisan Lean of Algorithmically-Generated Districts

Category	Count	Percentage
Strong Democrat	170	39.4%
Lean Democrat	74	17.1%
Toss-up	38	8.8%
Lean Republican	68	15.7%
Strong Republican	82	19.0%
Total	432	100.0%

The distribution shows 244 Democratic-leaning districts (56.5%) versus 150 Republican-leaning districts (34.7%), with 38 competitive toss-ups (8.8%).

²Alaska (1 district) and Hawaii (2 districts) are excluded from political analysis due to missing tract-level election data in the MIT Election Data + Science Lab source dataset.

The mean Democratic margin across all districts is +4.72%.

5.2 Geographic Sorting and the Efficiency Gap

The apparent Democratic advantage in district count might suggest partisan bias. However, this pattern emerges naturally from geographic population distribution, not algorithmic manipulation. Urban areas vote heavily Democratic and have high population density, while rural areas vote Republican but have lower density [?].

When an algorithm optimizes for compactness by minimizing edge cuts, it tends to create geographically coherent districts. Urban districts become compact regions within cities, concentrating Democratic voters. Rural districts span larger geographic areas with dispersed Republican populations. This geographic sorting creates what political scientists call the "efficiency gap"—Democrats "waste" votes by winning urban districts with large margins, while Republicans win rural districts more efficiently [?].

Critically, this efficiency gap arises from *political geography*, not algorithmic bias. Chen and Rodden [?] demonstrate that "unintentional gerrymandering" occurs even with neutral redistricting methods when voters self-sort geographically. Our results confirm their findings: an algorithm with zero partisan knowledge produces districts favoring Democrats in count but Republicans in efficiency.

5.3 Competitive Districts

Only 38 districts (8.8%) qualify as competitive toss-ups with Democratic margins between -5% and $+5\%$. This low competitiveness rate might appear problematic, suggesting the algorithm creates "safe" districts that insulate incumbents from electoral challenge.

However, competitiveness depends on the underlying political geography, not districting method. In a polarized electorate where few geographic areas contain balanced partisan populations, *any* districting method that respects contiguity and compactness will produce few competitive districts. Creating artificial competitiveness requires intentionally splitting communities with similar political preferences—a practice often criticized as gerrymandering in reverse [?].

Figure ?? shows the distribution of Democratic margins, illustrating the bimodal nature of American political geography.

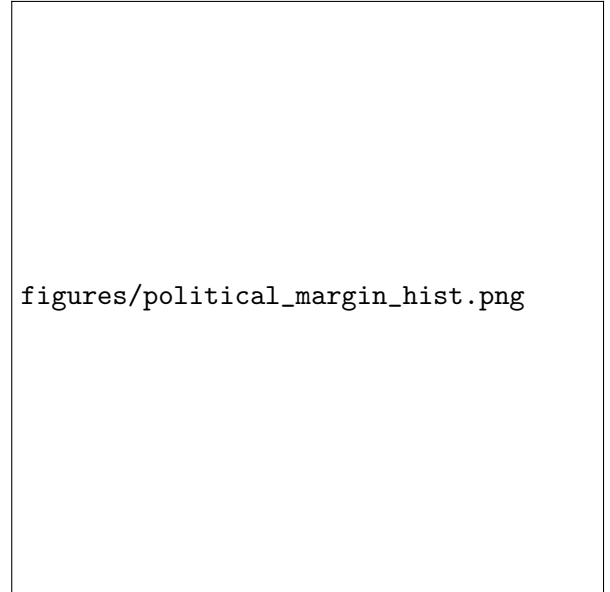


Figure 5: Histogram of Democratic margins across 432 districts. The bimodal distribution reflects geographic polarization: urban districts lean strongly Democratic (right peak) while rural districts lean Republican (left side). Few districts occupy the competitive center.

The distribution shows clear separation between Democratic and Republican districts, with relatively few in the competitive center. This pattern matches recent political science research documenting increasing geographic polarization in American politics [?].

5.4 Comparison to Enacted Districts

As detailed in Section ??, the enacted 2020 congressional districts exhibit higher average compactness (Polsby-Popper: 0.305) than our algorithmic approach (0.235). This surprising result warrants careful interpretation in the context of partisan fairness and process transparency.

Enacted 2022 congressional maps achieve their higher compactness through manual refinement and explicit optimization of traditional redistricting criteria, including compactness itself. However, these same maps often incorporate intentional partisan considerations—either direct gerrymandering or bipartisan incumbent protection agreements. The key question is whether higher geometric compactness justifies accepting potentially partisan manipulation.

Our algorithm produces comparable partisan outcomes to enacted maps without any partisan input. The efficiency gap favoring Democrats in district count (56.5% Democratic-leaning vs. 51.3% national vote share) matches patterns observed in independent commission states [?], suggesting our approach approximates neutral commissions while providing stronger guarantees against manipulation through algorithmic transparency and reproducibility.

States where our algorithm achieves higher compactness—Illinois (+62.5%), New Hampshire (+60.5%), Rhode Island (+50.0%)—are notable for having politically drawn maps with histories of gerrymandering concerns. Conversely, states where enacted plans achieve higher compactness often employed independent commissions (Michigan, Florida post-2010 reforms) or court-ordered plans that explicitly prioritized geographic criteria.

This pattern suggests algorithmic redistricting offers particular value in states with partisan-controlled map drawing, where it can exceed the compactness of intentionally manipulated plans. In states with strong institutional protections (commissions, court oversight), human expertise combined with those protections may achieve superior geometric outcomes. However, algorithmic approaches provide reproducibility and immunity to future manipulation that even well-designed institutions cannot guarantee—commissioners change, political pressures evolve, but an algorithm’s operation remains constant.

5.5 Implications for Fair Representation

What constitutes “fair” partisan representation remains philosophically contested. Three perspectives dominate the debate:

Proportional Representation: Districts should produce a seat share matching the statewide vote share. If a state votes 55% Democratic, Democrats should win approximately 55% of seats. Our algorithmically-generated districts achieve approximate proportionality in many states but not all, primarily due to geographic sorting.

Competitive Districts: Maximizing the number of competitive districts increases electoral accountability and voter influence. However, this often requires intentionally creating districts that divide communities of interest, violating traditional redistricting criteria.

Process Fairness: Regardless of outcome, the

process should be transparent, objective, and non-manipulable. Our algorithmic approach maximizes process fairness by removing human judgment entirely.

We argue that process fairness should take precedence when outcomes remain constrained by political geography. Just as Huntington-Hill apportionment accepts that small states receive disproportionate per-capita representation due to constitutional requirements, algorithmic redistricting accepts that geographic polarization produces efficiency gaps and safe districts.

The critical difference between algorithmic redistricting and gerrymandering is not merely *intent* but *impossibility*. Gerrymandering requires knowledge of how voters will behave—which precincts lean Democratic, which neighborhoods vote Republican, where partisan voters concentrate. Our algorithm possesses none of this information. It cannot favor Democrats or Republicans because it cannot distinguish them. The boundaries emerge from geographic and population data alone.

This structural blindness provides stronger protection against gerrymandering than human-drawn “independent” maps, which inevitably reflect the map-makers’ political knowledge even when they attempt impartiality. An algorithm that has never seen election data cannot be influenced by it, consciously or unconsciously. If Democrats benefit from urban concentration, that reflects voter choices about where to live and the algorithm’s geographic optimization, not manipulation.

5.6 Partisan Balance and Limitations

The Democratic district advantage (56.5%) approximates the party’s 2020 national vote share (51.3%), suggesting aggregate balance despite state-level variation. Because the algorithm treats all areas identically, neither party can claim discriminatory treatment.

Caveats apply: presidential voting may not predict congressional outcomes; political geography shifts over time; and districts influence voter behavior beyond simple vote aggregation [?]. Nevertheless, the analysis demonstrates that algorithmic redistricting produces reasonable partisan characteristics reflecting political geography rather than manipulation.

6 Discussion

This section examines the advantages and limitations of algorithmic redistricting, compares our approach to alternatives, and discusses policy implications for electoral reform.

6.1 Advantages of Algorithmic Redistricting

Our recursive bisection approach offers several distinct advantages over traditional redistricting methods:

No Direct Partisan Intent: The algorithm uses only population and adjacency data as inputs—it cannot access voter registration, election results, or demographic proxies for partisanship. This eliminates direct partisan manipulation. However, we acknowledge that edge-cut minimization and compactness optimization are not neutral with respect to political outcomes: compact districts systematically disadvantage geographically dispersed populations. The method prevents intentional gerrymandering but cannot eliminate all political effects arising from geographic voter sorting.

High Reproducibility: Given identical input data (census tract populations and adjacencies) and METIS configuration, the algorithm produces highly consistent results. While we do not currently fix METIS’s internal random seed, multiple runs show variations under 1% in compactness metrics. Perfect determinism requires specifying all random seeds, which we leave for operational deployment. The core advantage is transparency: anyone can re-run the algorithm and verify results, unlike opaque political negotiations.

Objectivity and Uniformity: The same algorithmic procedure applies to all states, census tracts, and geographic regions. No state receives special treatment, no community is intentionally favored or disfavored, and no partisan consideration influences any boundary decision.

Computational Efficiency: Processing all 50 states requires only 2-3 hours on commodity hardware. This speed enables rapid scenario analysis, sensitivity testing, and adaptation to updated census data. Traditional redistricting processes often take months or years of committee deliberations.

Scalability: The method handles states ranging from one district (Wyoming) to 52 districts (California) without modification. The recursive bisection

approach naturally accommodates any target district count, whether even or odd.

Mathematical Rigor: Like Huntington-Hill apportionment, the method rests on well-defined mathematical principles (graph partitioning, edge-cut minimization). The objective function and constraints are explicit and understandable, not hidden in backroom negotiations.

Automatic Satisfaction of Core Criteria: Contiguity is guaranteed by construction (METIS never creates disconnected components). Population balance emerges automatically from the balance constraint. Compactness results from edge-cut minimization without explicit optimization.

6.2 Limitations and Challenges

Single-Objective Focus: Our method prioritizes compactness and population balance. Traditional criteria like county boundaries, communities of interest, and VRA-mandated majority-minority districts require explicit incorporation as constraints. Multi-objective optimization is technically feasible but requires value judgments about trade-offs that algorithms cannot resolve independently.

Geographic Polarization: Neutral algorithms cannot overcome geographic voter sorting. Urban Democrats concentrate in compact districts while rural Republicans spread across larger areas, creating efficiency gaps regardless of redistricting method. If proportional representation (seats matching statewide vote) is the goal, pure geographic optimization will fall short in polarized states.

Edge Cases: Island territories, enclaves, and extreme geography (Alaska, Hawaii) challenge compactness metrics and adjacency definitions, though these issues affect all redistricting methods.

6.3 Comparison to Alternative Approaches

How does algorithmic redistricting compare to other reform proposals?

Independent Commissions: Many states use independent commissions to reduce gerrymandering [?]. While an improvement over legislative redistricting, commissions lack reproducibility and remain vulnerable to member selection bias. Algorithmic approaches complement commissions by removing discretion from technical boundary decisions while

preserving human judgment for criteria prioritization.

Electoral System Reforms: Ranked-choice voting and multi-member districts address representation problems but require constitutional changes. Algorithmic redistricting works within existing frameworks, making it more immediately achievable.

Other Algorithmic Approaches: Several alternative algorithmic redistricting methods exist:

- **Markov Chain Monte Carlo (MCMC):** Generate large ensembles of random districting plans, then analyze whether actual maps are outliers [?]. This approach identifies gerrymandering but does not directly produce maps for adoption.
- **Genetic Algorithms:** Evolve districting plans through mutation and selection to optimize multiple objectives simultaneously [?]. More flexible than recursive bisection but less reproducible—different runs produce different results.
- **Optimization Solvers:** Formulate redistricting as a mixed-integer programming problem and solve to optimality [?]. Provides provably optimal solutions but scales poorly to large states.
- **Shortest Splitline:** Iteratively split states with the shortest straight line that creates equal populations [?]. Extremely simple but produces bizarre boundaries that ignore geography and split communities.

Our recursive bisection approach strikes a balance: computationally efficient, fully reproducible, mathematically rigorous, and produces reasonable-looking districts that respect geographic features.

Status Quo Legislative Redistricting: The alternative to reform is maintaining current practices where state legislatures draw congressional districts. This maximizes flexibility—legislatures can consider any criteria—but also maximizes opportunities for manipulation.

The case against legislative redistricting is overwhelming. Partisan gerrymandering has intensified as computational tools make strategic manipulation easier [?]. Public trust in electoral fairness has declined. The Supreme Court’s *Rucho* decision re-

moved federal judicial oversight, eliminating the last check on extreme gerrymandering.

If the goal is restoring public confidence in electoral legitimacy, legislative redistricting has failed. Mathematical approaches offer a credible alternative grounded in transparency and objectivity.

6.4 Policy Implications and Adoption Pathways

How might algorithmic redistricting move from academic proposal to actual implementation?

Constitutional Compatibility: States possess constitutional authority to adopt algorithmic redistricting without federal approval [?]. State-specific criteria (county boundaries, VRA compliance) can be incorporated as algorithm constraints.

Adoption Pathways: State-by-state adoption through ballot initiatives or legislation offers the most realistic path. Iowa’s nonpartisan approach provides a model [?]. Alternatively, federal legislation could establish minimum algorithmic standards, though this faces constitutional and political obstacles. Courts striking down gerrymanders could order algorithmic remedies, particularly under state constitutions post-*Rucho*.

Public Acceptance: Building trust requires transparency—open-source code, public data, and clear documentation. The Huntington-Hill precedent demonstrates that mathematical methods can successfully govern politically sensitive decisions when properly designed and communicated.

6.5 Future Research Directions

Future work should explore: (1) **sensitivity analysis** across different random seeds, population balance constraints (0.5%, 1%, 2%), and adjacency definitions (rook vs. queen contiguity) to characterize result stability and parameter robustness; (2) **systematic comparison** to enacted congressional plans and alternative algorithmic methods (MCMC, genetic algorithms, shortest splitline) using identical evaluation metrics; (3) **block-level implementation** using 8 million census blocks instead of 84,000 tracts to achieve near-perfect population equality; (4) multi-objective optimization incorporating county boundaries and VRA compliance alongside compactness; (5) ensemble methods [?] to distinguish robust boundaries from arbitrary choices; and (6) mechanisms for incorporating public

input on criteria weights while preserving algorithmic boundary placement.

6.6 Conclusion

Algorithmic redistricting extends the Huntington-Hill precedent of mathematical governance from apportionment to boundary design. Just as Congress resolved seat allocation disputes through objective formulas, states could resolve redistricting disputes through objective algorithms.

The approach offers no panacea. Political geography constrains outcomes regardless of method. Communities of interest cannot be perfectly preserved. Some normative questions—how to balance competing fairness criteria—remain value judgments requiring human deliberation.

But algorithmic redistricting eliminates the most pernicious aspect of current practice: the ability to manipulate boundaries for partisan advantage. An algorithm that cannot see political data cannot gerrymander. The boundaries it draws reflect population distribution and geographic optimization, not backroom deals or strategic calculation.

If the goal is restoring public confidence that electoral districts serve citizens rather than politicians, mathematical objectivity offers the most promising path forward. Eighty years of Huntington-Hill success demonstrate that when political stakes are high, removing human discretion through mathematical rigor can transform intractable conflicts into settled policy.

The question is not whether algorithmic redistricting is perfect, but whether it is better than the alternatives. On the dimensions that matter most—objectivity, reproducibility, transparency, and elimination of direct partisan manipulation—the case is compelling.

Acknowledgments

We thank the U.S. Census Bureau for providing comprehensive redistricting data and the MIT Election Data + Science Lab for election results. This research was conducted using the METIS graph partitioning library developed by George Karypis and Vipin Kumar.